

Aquastainable project pitch

André van Heerden 241155

Open Window, Department of Fundamentals

INTERACTIVE DEVELOPMENT 300

Tsungai Katsuro

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A) Project Title

App Name: Aquastainable

Description: Intelligent aquarium ecosystem management, driven by sight and information.

B) Conceptualisation & Problem Statement

The Inspiration Card Connection:

- *Be More Creative:* The app avoids rigid forms, allowing users to express their setups visually and organically.
- *Camera:* The camera is the primary input tool. Users photograph their tanks, fish health issues, or chemical test strips to feed data directly into the system.
- *No Traditional UI Buttons:* To mimic the natural, fluid feel of looking into a water tank, the app completely eliminates standard rectangular buttons. It relies entirely on immersive gestures, contextual drags, and long-presses.

The Problem:

Beginner aquarium hobbyists face a steep learning curve. Failing to comprehend complex ecological cycles, volatile water chemistry, and safe ecosystem parameters often results in accidental fish loss, structural tank stress, and abandoned setups. Traditional tracking applications worsen this by requiring tedious manual data entry and form-filling, which alienates hobbyists.

How We Fix It :

Aquastainable eliminates the friction of traditional ecosystem tracking by converting visual data into actionable ecological intelligence. By leveraging a gesture-driven, buttonless user interface, users interact with their tanks naturally.

When a user captures a photo, our backend AI ecosystem analyzes the tank health, cross-references local town water diagnostics, and evaluates chemical parameters. It completely replaces guesswork with immediate, automated guidance on tank capacity, plant alignment, thermal baselines, and precise water change frequencies.

The Target Audience:

Aquarium hobbyists, aquascapers, and marine life enthusiasts who want an intelligent, seamless way to track their ecosystems without clumsy form-filling UI.

C) Core Features & Scope***User Roles:***

- **User:** The sole application role. Has full CRUD permissions over their own tank portfolios, livestock profiles, and chat logs. No administrative role is required within the client client application architecture.

Must-Have Features (MVP) :

- **Gesture-Driven Dashboard:** Visual card management interface completely devoid of traditional buttons, utilizing drag-and-drop and long-press actions to navigate.

- **Visual Tank Tracking:** Industrial tracking of aquariums using periodic photographic history logs instead of manual metrics text boxes.
- **Biometric & Volume Profiling:** Tank dimension/size capture mapped against user livestock additions (specifying fish count, individual profiles, or school clusters).
- **Location-Based Water Diagnostics:** Town-level geolocation configuration to scan for localized regional water anomalies or municipal supply issues.
- **AI Ecosystem Advisory Engine:** Automated calculation of maximum biological loading capacity, optimal temperature ceilings, compatible flora (plants), and required water alteration frequencies.
- **AI Visual Consultation Chamber:** A dedicated interactive support section where users upload photos alongside raw text inquiries to receive contextual diagnostic feedback.

Nice-to-Have Features (Future Considerations) :

- **Push Notification Reminders:** Micro-interactions nudging the user to perform scheduled water changes or parameter checks based on AI calculations.
- **Test Strip Color Matching Engine:** Automated computer vision analysis that parses test strip colors directly from a camera snap to extract exact pH, Ammonia, and Nitrate values.
- **Community Showcase Feed:** A view-only structural layout where users can share their visual tank progress cards anonymously with other local hobbyists.

SMART Objectives :

- **Specific:** Plan and build the cross-platform mobile app *Aquastainable* using React Native and Firebase, implementing a 100% buttonless UI driven entirely by touch gestures (long-press, drag-and-drop, swipes).
- **Measurable:** Successfully implement full CRUD functionality for a single user role across 4 core database entities (Users, Aquariums, Livestock, and Consultations), alongside a fully functional camera feature that uploads photos to Firebase Storage.
- **Achievable:** The scope is contained to one user role and leverages established production libraries to manage the specialized UI.
- **Relevant:** Directly fulfills the randomized Inspiration Card constraints (Camera, Geolocation, and No Traditional UI Buttons) to solve the core problem of beginner aquarium hobbyists.
- **Time-bound:** To be entirely developed, tested, and finalized over a strict 4-week timeline running from Week 3 to the Week 7 submission deadline.

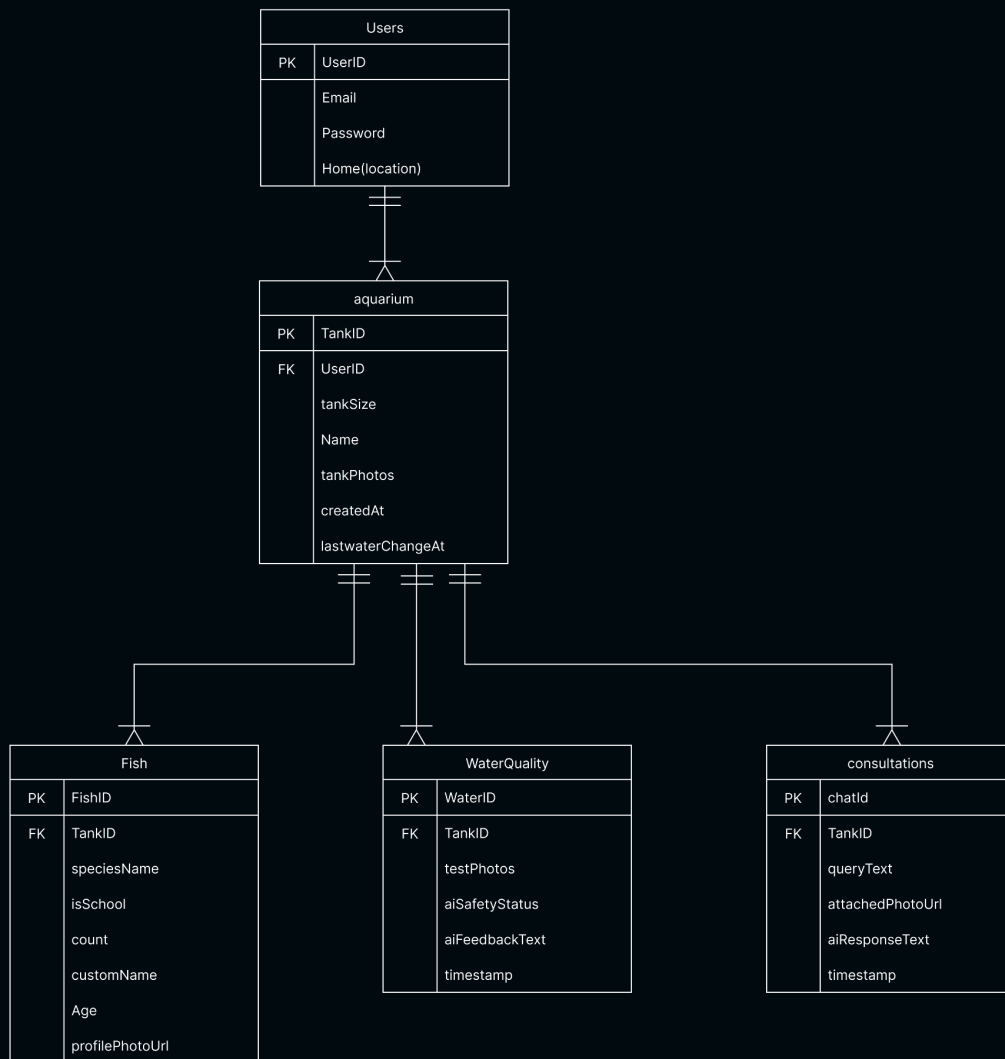
D) Technical Planning & Architecture

Technologies & Frameworks

- **React Native:** Cross-platform mobile development framework allowing a single codebase for Android and iOS.
- **TypeScript:** Strongly-typed programming language that scales cleanly and mitigates runtime bugs.
- **Google Firebase Auth:** Out-of-the-box secure user authentication and session management.

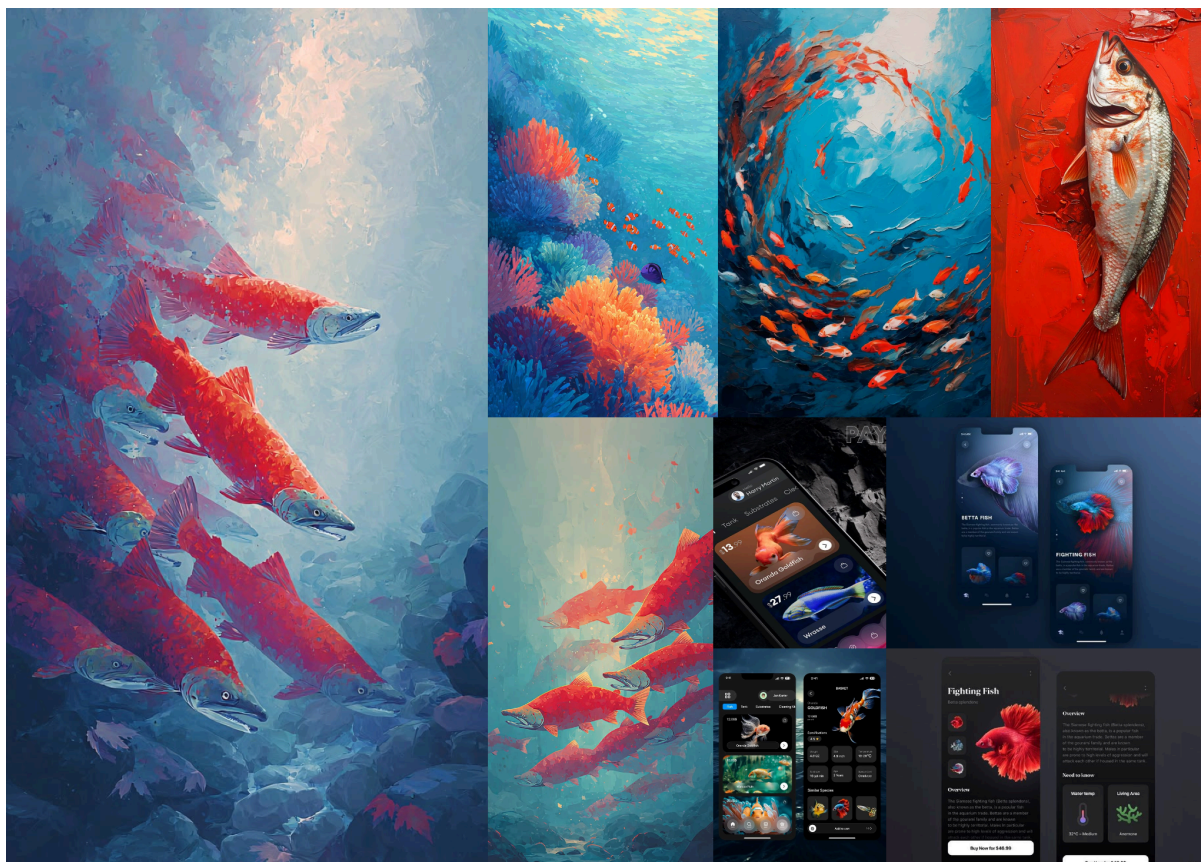
- **Google Cloud Firestore:** Scalable, real-time NoSQL document database used to store flexible structural data model profiles.
- **Google Cloud Storage:** Secure binary large object (BLOB) storage optimized for hosting user tank and fish photographs.
- **React Native Gesture Handler:** Native-driven touch and gesture tracking library to facilitate buttonless UI interactions.
- **React Native Reanimated:** High-performance animation engine utilized to manage fluid UI transitions and component tweening.
- **Expo Camera / React Native Vision Camera:** Low-level camera API integration providing raw image capture capabilities for backend processing.

Entity-Relationship Diagram (ERD)



E) UI/UX Considerations

Moodboard



Style guide

UI Style Guidelines

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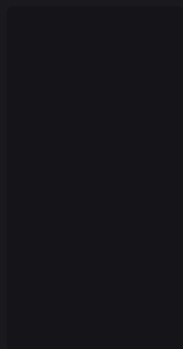
- 01. Colors
- 02. Typography
- 03. Logo

01. Colors

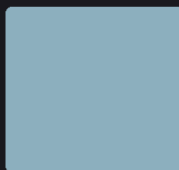
Brand Colors



Primary



Secondary



complementary



complementary

State Colors



Info
#2F80ED



Success
#37F611



Warning
#FFB01F



Error
#EB5757

02. Typography

Montserrat

Google Fonts

Aa

Heading

Line height and paragraph
spacing for heading is
1.1 x font size

Letter spacing: 10%

H1 - Montserrat - 22pt
Bold

H2 - Montserrat - 20pt
Bold

H3 - Montserrat - 18pt Semi
Bold

H4 - Montserrat - 17pt Semi
Bold

Body - Montserrat - 15pt medium

Caption - Montserrat - 13pt medium

03. Logo



F) Project Roadmap

Week 3 - 4: Frontend Development & Gesture Engine Foundations:

- Project bootstrap with React Native & TypeScript
- Build out buttonless, gesture-controlled navigation screens
- Implement Camera access layouts and fluid micro-interaction states

Week 5: Backend Architectural Integration:

- Link Firebase Auth workflows for safe user onboarding
- Connect file upload flows linking client-side camera to Cloud Storage
- Wire core API gateway triggers out to the external AI processing node

Week 6: NoSQL Database Pipeline & Feature Implementation

- Implement Firestore document modeling for tanks, fish, and chats
- Build out absolute CRUD functionalities via gestures (delete/update)
- Wire up the AI consultation chat feed with real-time payload updates

Week 7: Staging, Security Rules, Optimization & Deployment

- Apply strict security rule logic validations inside Firebase consoles
- Optimise cross-platform performance bugs on Android & iOS layers
- Compile release builds for Google Play Console & Apple App Store